

Pricing and hedging of financial derivatives — Peter Tankov 2022–2023, main session

No documents/calculators allowed, except for 1 sheet of paper (A4, two-sided) with handwritten notes. Exam length 2 hours

Exercise 1: stochastic dividends

Consider a financial market with two risky assets: a dividend-paying stock with price denoted by S_t and a forward contract on this stock, with maturity T , as well as a risk-free asset with zero interest rate. To simplify notation, we assume that the forward contract has pay-off equal exactly to S_T at time T . We denote the time- t price of this forward contract by F_t . The future dividends paid by the stock are unknown, and we assume that the *dividend rate* is given by the stochastic process q_t . This means that during every small interval dt , the stock pays dividends $S_t q_t dt$. Since the future dividends are unknown there is no direct relationship between the forward price and the spot price, and we need to model separately the two processes. We assume that under the historical measure $\mathbb{P}$, the stock price and the forward price follow the dynamics

$$\begin{aligned}\frac{dS_t}{S_t} &= \mu_t^S dt + \sigma^S(t) dW_t \\ \frac{dF_t}{F_t} &= \mu_t^F dt + \sigma^F(t) (\rho dW_t + \sqrt{1 - \rho^2} dW'_t),\end{aligned}$$

where (W, W') is a standard 2-dimensional Brownian motion (with independent components) and $\sigma^S(t)$ and $\sigma^F(t)$ are deterministic volatilities.

1. Since the stock pays dividends, a portfolio containing a single stock at all times is not self-financing. We shall therefore, consider the value X_t of a portfolio, which contains one unit of stock at time $t = 0$, and where all dividends paid by the stock are reinvested into the same stock. X is called the cum-dividend process. We admit that $X_t = S_t e^{\int_0^t q_s ds}$. Deduce the dynamics of X_t under the historical measure $\mathbb{P}$.
2. We now consider the market with two risky assets: the forward contract F_t and the cum-dividend process X_t . Write down the two-dimensional risk premium vector and the 2x2 volatility matrix for this model. Determine the density of the risk-neutral measure $\mathbb{Q}$ and write down the dynamics of F_t and X_t under this measure. We assume that the risk premium vector is such that the risk-neutral measure is well defined. Deduce the dynamics of S_t under the risk-neutral measure.
3. In this question, we consider the so called dividend swap: a financial product with pay-off $e^{\int_0^T q_t dt}$, used to hedge the risk of future dividends. Compute the price D_t of the dividend swap at time $t < T$.
Hint: use the representation $e^{\int_0^T q_t dt} = \frac{X_T}{F_T}$ and the risk-neutral pricing formula.
4. Using Itô formula, determine the dynamics of D_t .
5. Compute the price at time t of the “dividend option” with pay-off at time T given by $H_T = \left(e^{\int_0^T q_t dt} - K\right)^+$.
Hint: consider this option as an option on D_T , then use the dynamics of D_t .
6. Determine the hedging strategy for the dividend option of the previous question.

Exercise 2: gamma swap in Heston stochastic volatility model

A gamma swap is a modification of the variance swap where the variance of the underlying is weighted by the underlying's value. The pay-off (using the continuous approximation) is thus given by

$$H_T^G = \frac{1}{T} \int_0^T \frac{S_t}{S_0} \sigma_t^2 dt.$$

Remark that we have dropped the constant leg of the swap to simplify the formulas. We also recall the pay-off of the forward variance swap, where we similarly drop the fixed leg:

$$H_{T_1, T_2} = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \sigma_t^2 dt.$$

In this exercise we suppose that the **interest rate is zero** and the underlying asset S follows the Heston stochastic volatility model under the **risk-neutral** probability:

$$\frac{dS_t}{S_t} = \sigma_t dW_t, \quad dv_t = \kappa(\theta - v_t)dt + \delta\sqrt{v_t}(\rho dW_s + \sqrt{1 - \rho^2} dW'_s), \quad v_t = \sigma_t^2, \quad (1)$$

where (W, W') is a two-dimensional standard Brownian motion (with independent components).

1. Compute the expectation $\mathbb{E}[v_T | \mathcal{F}_t]$. Hint: use Itô formula to show that

$$v_T = \theta + (v_t - \theta)e^{-k(T-t)} + \int_t^T e^{-k(T-s)} \delta\sqrt{v_s}(\rho dW_t + \sqrt{1 - \rho^2} dW'_t),$$

then take the conditional expectation. You may assume without proof that the stochastic integrals have zero expectation.

2. From the previous question, deduce the price at time $t < T_1$ of the forward variance swap with pay-off H_{T_1, T_2} in Heston's model, denoted by $F_t(T_1, T_2)$.
3. We admit that $\mathbb{E}[S_T] = S_0$. Introduce the probability $\tilde{\mathbb{Q}}$ defined by $\frac{d\tilde{\mathbb{Q}}}{d\mathbb{Q}}|_{\mathcal{F}_T} = \frac{S_T}{S_0}$. Show that under $\tilde{\mathbb{Q}}$, the process v follows the same dynamics as in (1), with different parameters $\tilde{\kappa}, \tilde{\theta}, \tilde{\delta}$ to be determined. You may assume that $\kappa > \delta\rho$.
4. Using the result of the previous question, compute the price of the gamma swap at time $t = 0$, denoted by $\Gamma_0(T)$.

We are now interested in hedging the gamma swap.

5. Compute the price of the gamma swap at date $t \in (0, T)$. Hint: the variance accumulated until date t is $\mathcal{F}_t$ -measurable and must therefore be taken out of the expectation.
6. Applying the Itô formula, compute the dynamics of the gamma swap.
7. Compute the dynamics of the forward variance swap.
8. From the two previous questions, deduce a strategy for hedging the gamma swap with a forward variance swap and the underlying.

Solution

Exercise 1: stochastic dividends

1. The portfolio contains at time t the quantity $\delta_t = \frac{X_t}{S_t}$ stocks. Every stock pays the amount $S_t q_t dt$ dividends during a small interval dt . Therefore, all stocks in the portfolio pay the amount $\delta_t S_t q_t dt$, which can be used to purchase $\delta_t q_t dt$ shares. Therefore, $d\delta_t = \delta_t q_t dt$, and $\delta_t = e^{\int_0^t q_s ds}$. We can therefore write

$$\frac{dX_t}{X_t} = (q_t + \mu_t^S)dt + \sigma^S(t)dW_t.$$

2. Introducing the risk premium vector

$$\lambda_t = \Sigma(t)^{-1} \begin{pmatrix} q_t + \mu_t^S - r \\ \mu_t^F - r \end{pmatrix}, \quad \Sigma(t) = \begin{pmatrix} \sigma^S(t) & 0 \\ \sigma^F(t)\rho & \sigma^F(t)\sqrt{1-\rho^2} \end{pmatrix}$$

we write:

$$d \begin{pmatrix} X_t \\ F_t \end{pmatrix} = \text{diag}(X_t, F_t) \left\{ r \mathbf{1} dt + \Sigma(t)(\lambda_t dt + d \begin{pmatrix} W_t \\ W'_t \end{pmatrix}) \right\}$$

The risk-neutral probability is defined by

$$\frac{d\mathbb{Q}}{d\mathbb{P}}|_{\mathcal{F}_T} = e^{-\int_0^T (\lambda_s^1 dW_s + \lambda_s^2 dW'_s) - \frac{1}{2} \int_0^T |\lambda_s|^2 ds}$$

Let $\widetilde{W}_t = W_t + \int_0^t \lambda_s^1 ds$ and $\widetilde{W}'_t = W'_t + \int_0^t \lambda_s^2 ds$. Then $(\widetilde{W}, \widetilde{W}')$ is a $\mathbb{Q}$ -Brownian motion and we can write:

$$\begin{aligned} \frac{dX_t}{X_t} &= rdt + \sigma^S(t)d\widetilde{W}_t \\ \frac{dF_t}{F_t} &= rdt + \sigma^F(t)(\rho d\widetilde{W}_t + \sqrt{1-\rho^2}d\widetilde{W}'_t), \end{aligned}$$

and consequently

$$\frac{dS_t}{S_t} = (r - q_t)dt + \sigma^S(t)d\widetilde{W}_t$$

3. The price of the dividend swap is given by

$$\begin{aligned} D_t &= e^{-r(T-t)} \mathbb{E} \left[\frac{X_T}{F_T} \middle| \mathcal{F}_t \right] \\ &= e^{-r(T-t)} \frac{X_t}{F_t} \mathbb{E} \left[e^{\int_t^T \sigma^S(s) d\widetilde{W}_s - \rho \int_t^T \sigma^F(s) d\widetilde{W}_s - \sqrt{1-\rho^2} \int_t^T \sigma^F(s) d\widetilde{W}'_s - \frac{1}{2} \int_t^T \sigma^S(s)^2 ds + \frac{1}{2} \int_t^T \sigma^F(s)^2 ds} \right] \\ &= e^{-r(T-t)} \frac{X_t}{F_t} e^{\frac{1}{2} \int_t^T (\sigma^S(s) - \rho \sigma^F(s))^2 ds + \frac{1}{2} \int_t^T (1-\rho^2) \sigma^F(s)^2 ds - \frac{1}{2} \int_t^T \sigma^S(s)^2 ds + \frac{1}{2} \int_t^T \sigma^F(s)^2 ds} \\ &= e^{-r(T-t)} \frac{X_t}{F_t} e^{-\int_t^T \rho \sigma^S(s) \sigma^F(s) ds + \frac{1}{2} \int_t^T \sigma^F(s)^2 ds} \end{aligned}$$

4. Itô formula yields:

$$\begin{aligned} \frac{dD_t}{D_t} &= rdt + \frac{dX_t}{X_t} - \frac{dF_t}{F_t} + \rho \sigma^S(t) \sigma^F(t) dt - \frac{1}{2} \sigma^F(t)^2 dt - \frac{d\langle X, F \rangle_t}{X_t F_t} + \frac{d\langle F \rangle_t}{F_t^2} \\ &= rdt + \frac{dX_t}{X_t} - \frac{dF_t}{F_t} \\ &= rdt + \sigma^S(t) d\widetilde{W}_t - \sigma^F(t) (\rho d\widetilde{W}_t + \sqrt{1-\rho^2} d\widetilde{W}'_t), \end{aligned}$$

By the Black-Scholes formula we then get that the price of the option, denoted by P_t is given by

$$P_t = D_t N(d_1) - K e^{-r(T-t)} N(d_2),$$

$$d_{12} = \frac{\log \frac{D_t}{K e^{-r(T-t)}} \pm \frac{1}{2} V_t^T}{\sqrt{V_t^T}}$$

$$V_t^T = \int_t^T (\sigma^S(s)^2 + \sigma^F(s)^2 - 2\rho\sigma^S(s)\sigma^F(s)) ds$$

5. By Itô formula

$$dP_t = N(d_1) dD_t - K e^{-r(T-t)} N(d_2) r dt.$$

Substituting the dynamics of D_t , we get:

$$dP_t = N(d_1) D_t \left(\frac{dX_t}{X_t} - \frac{dF_t}{F_t} \right) + P_t r dt.$$

The hedging portfolio will therefore go long $\frac{N(d_1)D_t}{X_t}$ units of the cum-dividend portfolio, go $\frac{N(d_1)D_t}{F_t}$ of the forward contract and some risk-free asset to make the portfolio self-financing.

Exercise 2: gamma swap in Heston stochastic volatility model

1. From the hint, we deduce that

$$\mathbb{E}[v_T | \mathcal{F}_t] = \theta + (v_t - \theta) e^{-k(T-t)}.$$

2. By the risk-neutral pricing formula,

$$\begin{aligned} F_t(T_1, T_2) &= \mathbb{E}[H_{T_1, T_2} | \mathcal{F}_t] = \mathbb{E}\left[\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \sigma_s^2 ds \middle| \mathcal{F}_t\right] = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \mathbb{E}[\sigma_s^2 | \mathcal{F}_t] ds \\ &= \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} (\theta + (v_t - \theta) e^{-k(s-t)}) ds = \theta + \frac{(v_t - \theta)}{T_2 - T_1} \int_{T_1}^{T_2} e^{-k(s-t)} ds \\ &= \theta + \frac{(v_t - \theta)}{T_2 - T_1} \frac{e^{-k(T_1-t)} - e^{-k(T_2-t)}}{k} \end{aligned}$$

3. By Girsanov theorem,

$$\widetilde{W}_t = W_t - \int_0^t \sigma_s ds$$

is a $\widetilde{\mathbb{Q}}$ -Brownian motion, so that

$$dv_t = \kappa \theta dt - (\kappa - \delta \rho) v_t dt + \delta \sqrt{v_t} (\rho d\widetilde{W}_t + \sqrt{1 - \rho^2} dW'_t)$$

Thus the new parameters are given by $\tilde{\kappa} = \kappa - \delta \rho$ and $\tilde{\theta} = \frac{\kappa \theta}{\kappa - \delta \rho}$.

4. Using the formula for the variance swap under the new probability, with $T_1 = 0$ and $T_2 = T$, we get

$$\Gamma_0(T) = \tilde{\theta} + \frac{(v_0 - \theta')}{T} \frac{1 - e^{-\tilde{k}T}}{\tilde{k}}$$

5. By the risk-neutral pricing formula,

$$\begin{aligned}
\Gamma_t(T) &= \frac{1}{T} \int_0^t \frac{S_s \sigma_s^2}{S_0} ds + \mathbb{E} \left[\frac{1}{T} \int_t^T \frac{S_s \sigma_s^2}{S_0} ds \middle| \mathcal{F}_t \right] \\
&= \frac{1}{T} \int_0^t \frac{S_s \sigma_s^2}{S_0} ds + \frac{S_t}{S_0} \frac{T-t}{T} \tilde{\mathbb{E}} \left[\frac{1}{T-t} \int_t^T \sigma_s^2 ds \middle| \mathcal{F}_t \right] \\
&= \frac{1}{T} \int_0^t \frac{S_s \sigma_s^2}{S_0} ds + \frac{S_t}{S_0} \frac{T-t}{T} \left\{ \tilde{\theta} + \frac{(v_t - \tilde{\theta})}{T-t} \frac{1 - e^{-\tilde{k}(T-t)}}{\tilde{k}} \right\} \\
&= \frac{1}{T} \int_0^t \frac{S_s \sigma_s^2}{S_0} ds + \frac{S_t}{S_0} \frac{T-t}{T} \tilde{\theta} + \frac{S_t}{S_0} \frac{(v_t - \tilde{\theta})}{T} \frac{1 - e^{-\tilde{k}(T-t)}}{\tilde{k}}
\end{aligned}$$

6. By Itô formula,

$$d\Gamma_t(T) = \frac{dS_t}{S_0} \frac{T-t}{T} \tilde{\theta} + \frac{dS_t}{S_0} \frac{(v_t - \tilde{\theta})}{T} \frac{1 - e^{-\tilde{k}(T-t)}}{\tilde{k}} + \frac{S_t}{S_0} \frac{dv_t - \kappa(\theta - v_t)dt}{T} \frac{1 - e^{-\tilde{k}(T-t)}}{\tilde{k}}$$

7. By Itô formula,

$$dF_t(T_1, T_2) = \frac{e^{-\kappa(T_1-t)} - e^{-\kappa(T_2-t)}}{\kappa(T_2 - T_1)} \{dv_t - \kappa(\theta - v_t)dt\}$$

8. From the two previous questions, we deduce:

$$d\Gamma_t(T) = \left\{ \frac{T-t}{S_0 T} \tilde{\theta} + \frac{(v_t - \tilde{\theta})}{S_0 T} \frac{1 - e^{-\tilde{k}(T-t)}}{\tilde{k}} \right\} dS_t + \frac{S_t}{S_0 T} \frac{1 - e^{-\tilde{k}(T-t)}}{\tilde{k}} \frac{\kappa(T_2 - T_1)}{e^{-\kappa(T_1-t)} - e^{-\kappa(T_2-t)}} dF_t(T_1, T_2)$$